

GO / NO GO

Intelligent Sea Breeze Prediction System
for Water Sports Using Machine Learning

Roger Defez

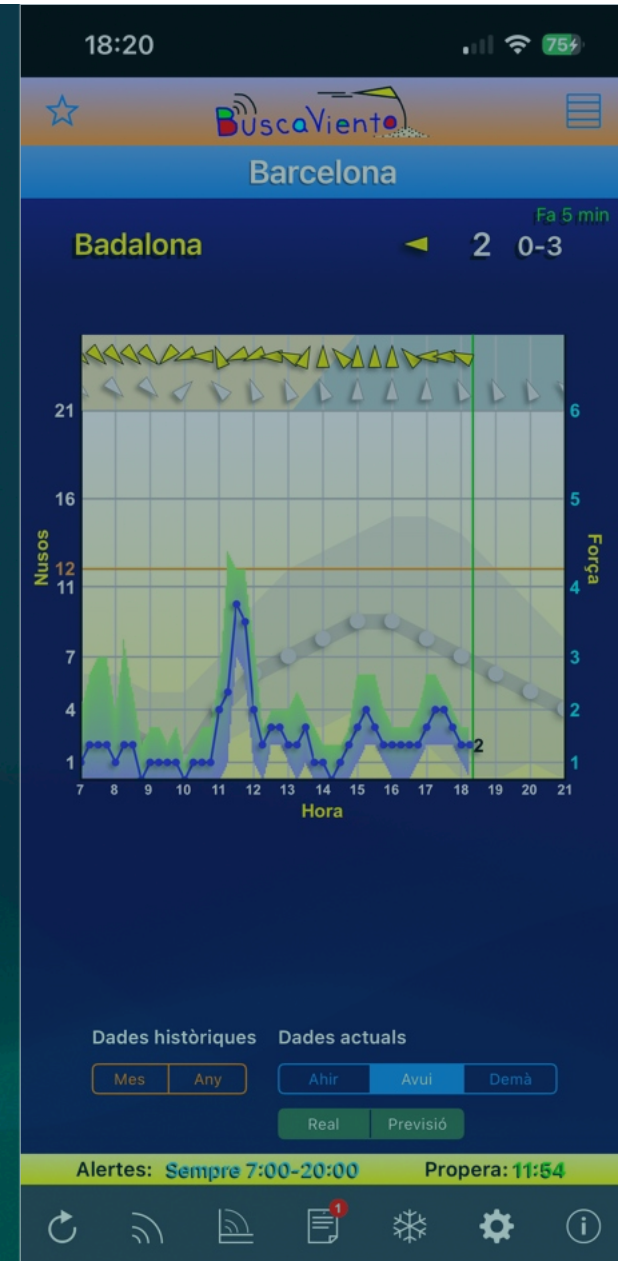
Bootcamp Data Analyst

2026 - 06 - 18

Terminology

Explanation of main variables and terminology of the dataset

- Unreliable forecasts
- Local variability
- Travel decision
- Wasted time



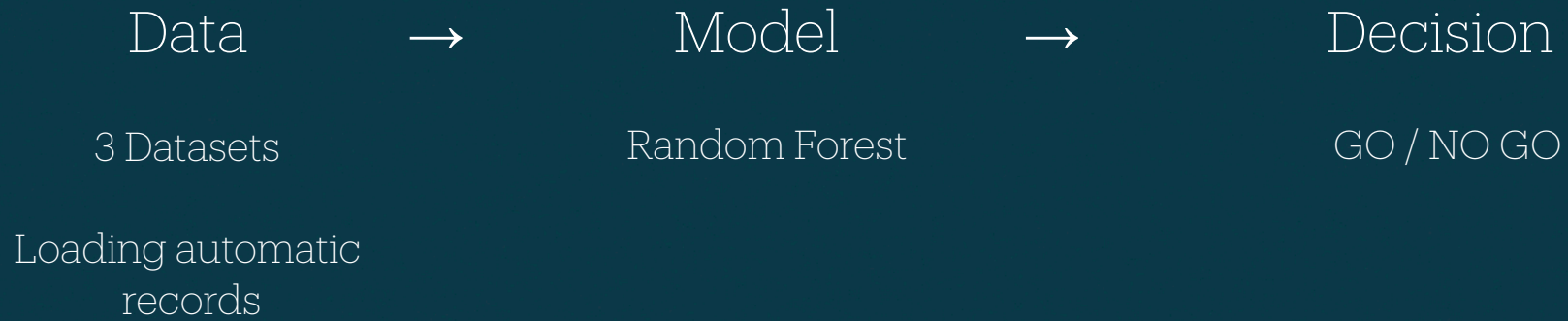
The Problem

How can you know in advance if there will be a good session on the water?

- Unreliable forecasts
- Local variability
- Travel decision
- Wasted time



Project Objective



- Detect favorable conditions
- Identify Sea Breeze events
- Generate GO / NO GO recommendation

Data Sources

2010-2025 hourly weather and marine data

- Three real Datasets
- More than 4 million records
- Data pipeline in 5 steps

Buscaviento
BDLONA 2010-2025
Viento spot ~15 min

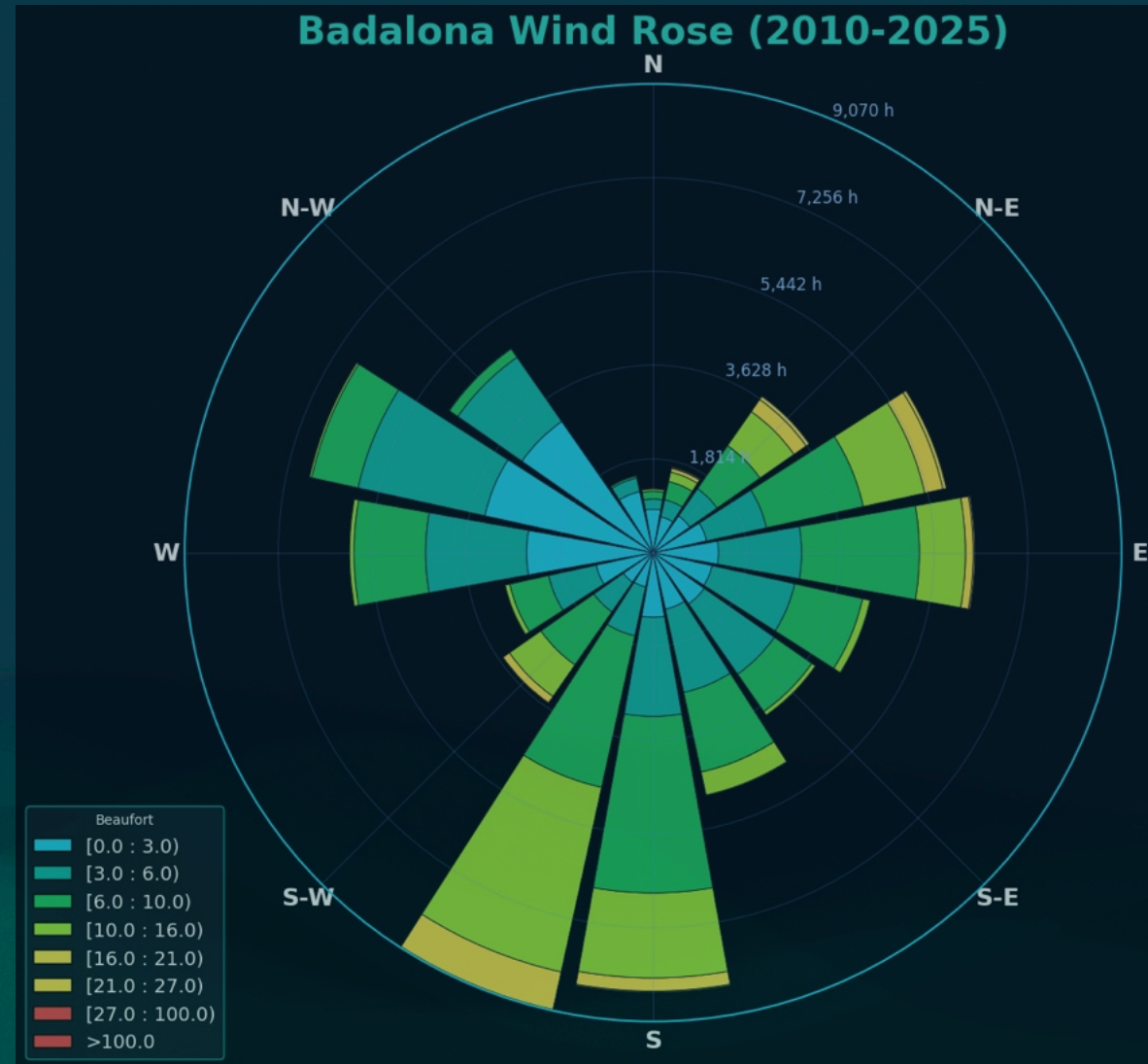
XEMA Museu Badalona
Meteo horaria 24h

Boya Barcelona
SST + oleaje horario

Merge horario → Features diarias → Random Forest → GO / NO GO

The datasets

- Buscaviento Badalona · wind spot · 2010-2025 · Granularity 15 minutes
- XEMA Museu Badalona · Meteocat · Atmospheric data 2010-2025 · Granularity 1h
- Oceanographic buoy Barcelona · Puertos del Estado · Sea Surface Temperature (SST) 2010-2025 · Granularity 1h

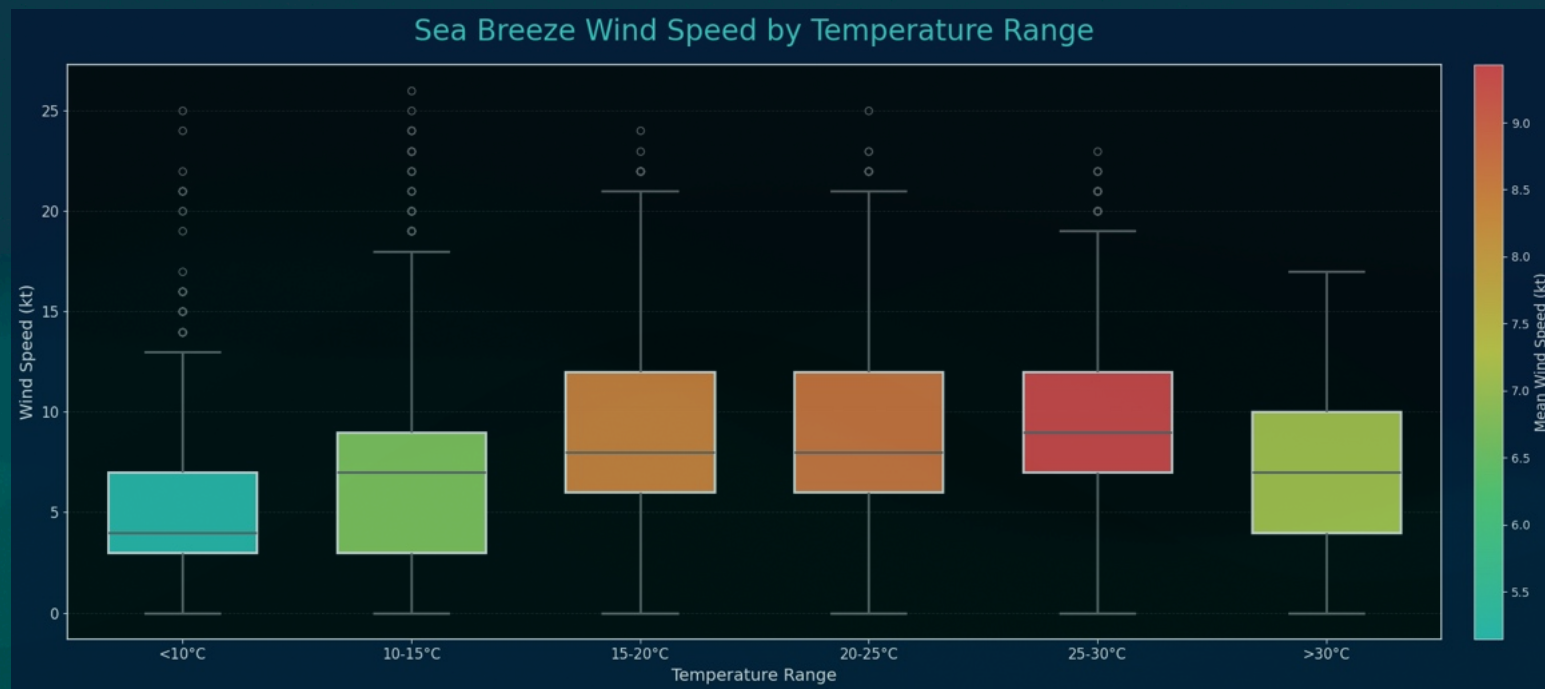


Buscaviento Badalona · wind spot · 2010-2025

Hypothesis

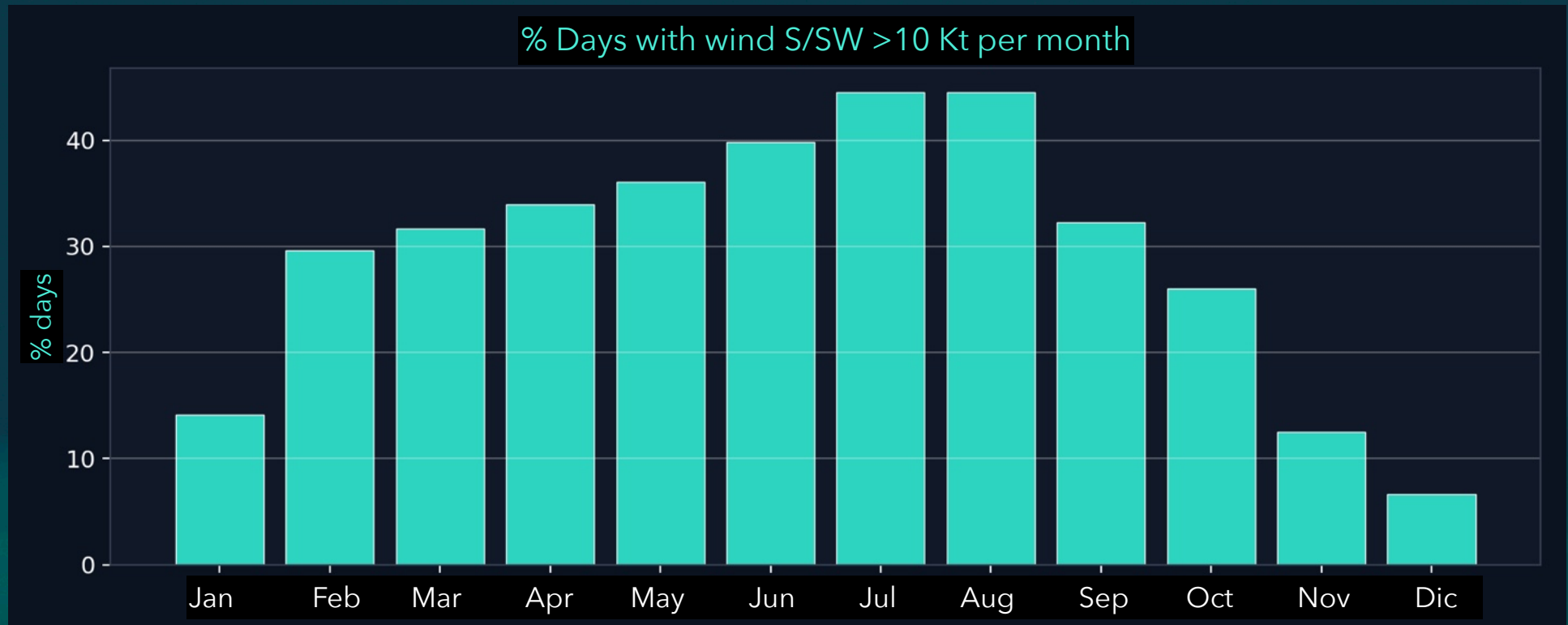
What would explain the entry of Sea Breeze Wind?

- More sun → more Sea Breeze Wind
- More land-sea contrast → more Sea Breeze Wind
- Greater day-night range → more Sea Breeze Wind



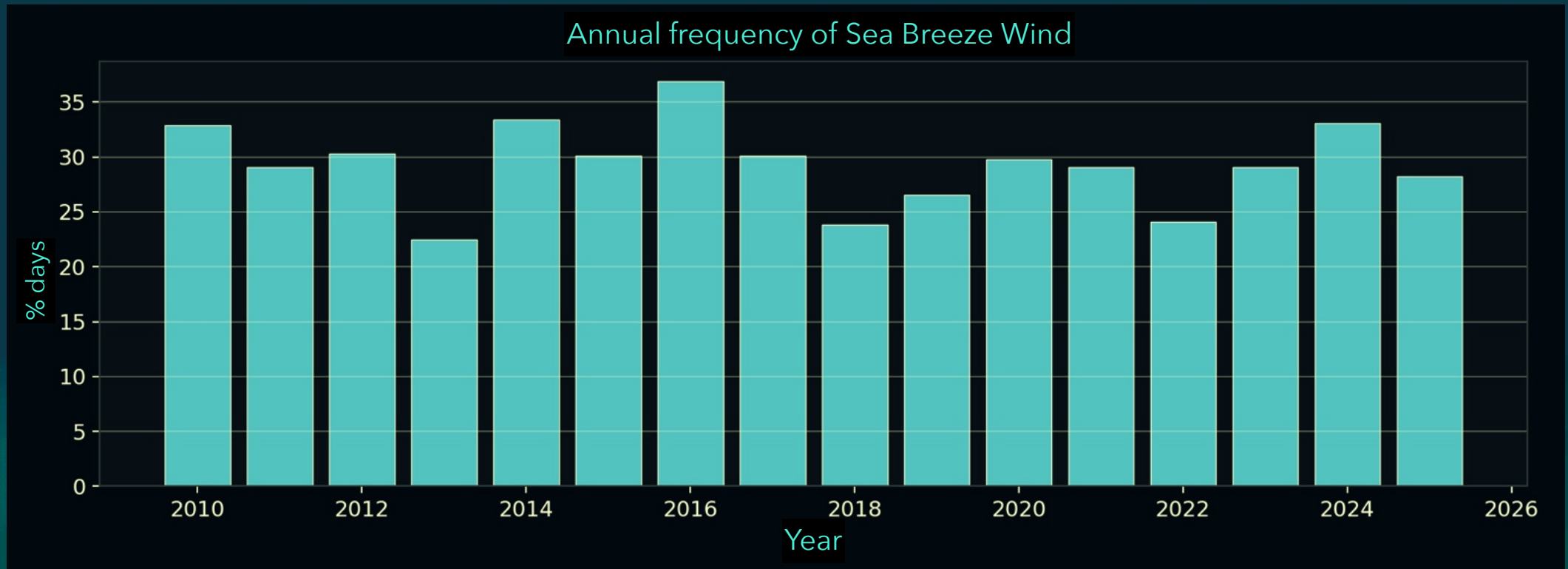
Seasonal pattern

The warmer months have a higher percentage of days with Sea Breeze Wind



Long term wind stability

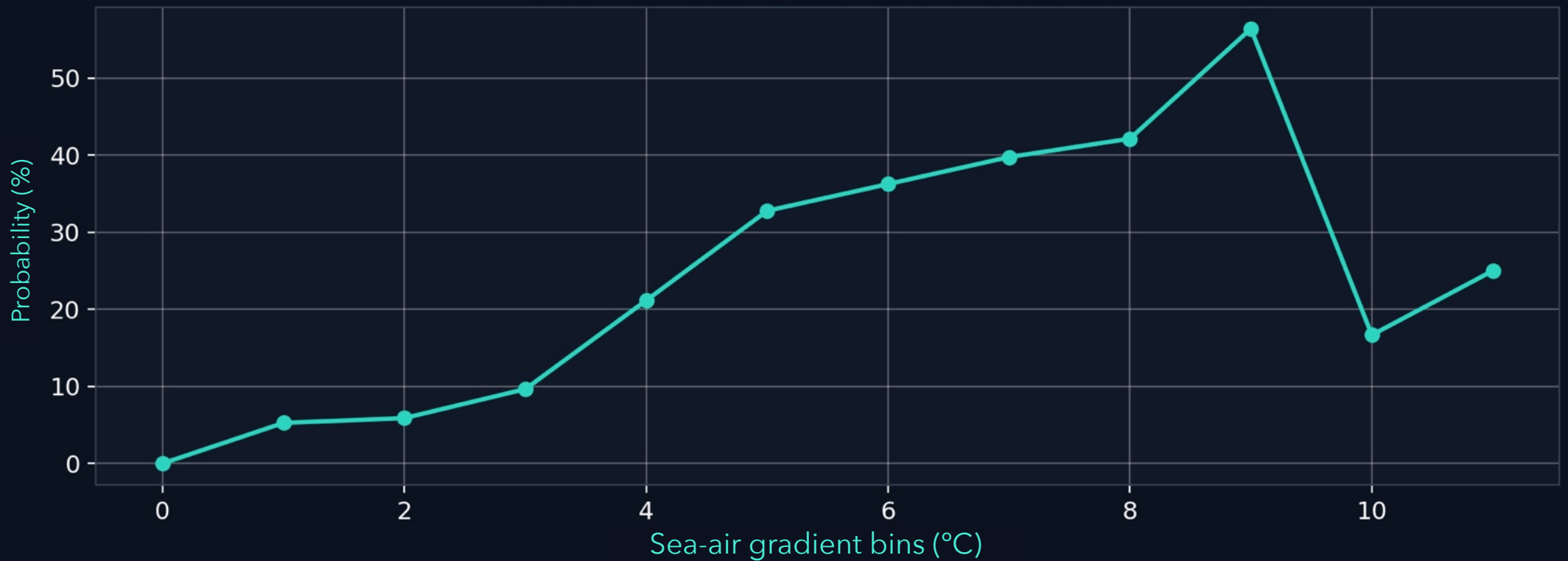
There is some interannual stability in the number of days with Sea Breeze Wind



Hypothesis validation

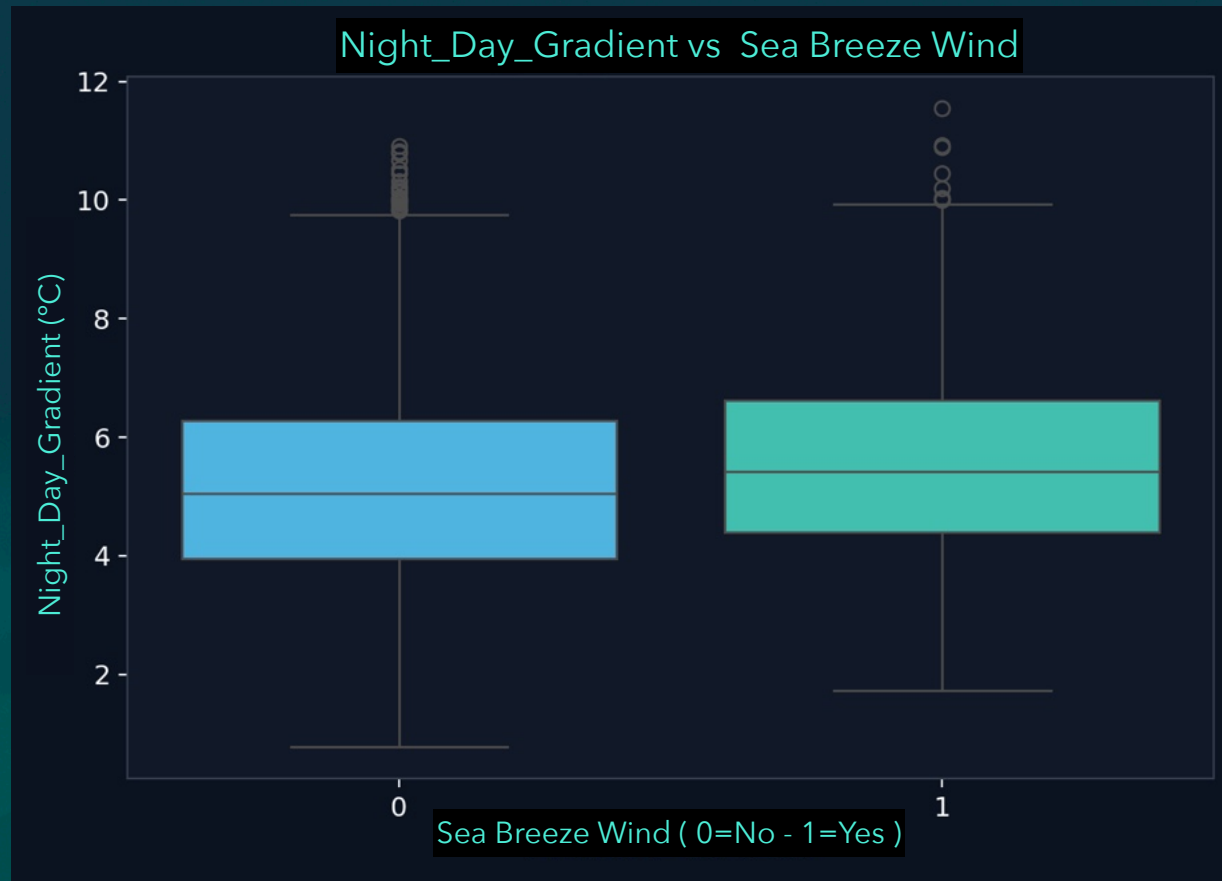
More land-sea contrast $\rightarrow$ more Sea Breeze Wind

Probability of Sea Breeze Wind vs higher daily sea-land thermal contrast



Hypothesis validation

More night-day contrast $\rightarrow$ more Sea Breeze Wind



Season × Month

How the seasons and months affect the appearance of Sea Breeze Wind?



Complete Random Forest Analysis

Temporal Split Strategy

Temporary separation prevents information leaks and simulates real production behavior.

- 9 variables · 5.806 days
- Train → 2010-2021
- Test → 2022-2023
- Validation → 2024-2025

Complete Random Forest Analysis

Relevant variables to analyze

****Features for the Random Forest**** (without observed wind columns → we avoid direct leakage):

- Night_Day_Gradient
- Irradiancia_solar_max
- Air_See_Gradient_max
- pressure_mean
- precip_total
- sst_mean
- month
- day_of_year
- season_code

Baseline Analysis

DummyClassifier always predicts the most frequent class.

Result: $F1=0$ for Sea Breeze Wind days.

Conclusion: any subsequent improvement reflects real learning.

≡ BASELINE ≡					
		precision	recall	f1-score	support
	0	0.694	1.000	0.819	507
	1	0.000	0.000	0.000	224
accuracy				0.694	731
macro avg		0.347	0.500	0.410	731
weighted avg		0.481	0.694	0.568	731

Random Forest Architecture

- 500 decision trees.
- Maximum depth: 12.
- Class balancing activated.
- Advantages: robustness, non-linear relationships and good interpretability.

Results Interpretation

The model detects approximately 62% of actual thermal events.

- Accuracy = 0.648
- Precision = 0.447
- Recall = 0.616
- F1 = 0.518
- ROC-AUC = 0.714

```
≡≡≡ RANDOM FOREST – TEST ≡≡≡
              precision    recall  f1-score   support

         0       0.796       0.663       0.723        507
         1       0.447       0.616       0.518        224

    accuracy                   0.648        731
   macro avg       0.621       0.639       0.621        731
  weighted avg       0.689       0.648       0.660        731

F1:   0.518
AUC:  0.714
```

Confusion Matrix

Desde un punto de vista operativo es preferible una falsa alarma a perder un día excepcional de navegación.

- True Negative - **TN** = 336
- False Positive - **FP** = 171
- False Negative - **FN** = 86
- True Positive - **TP** = 138

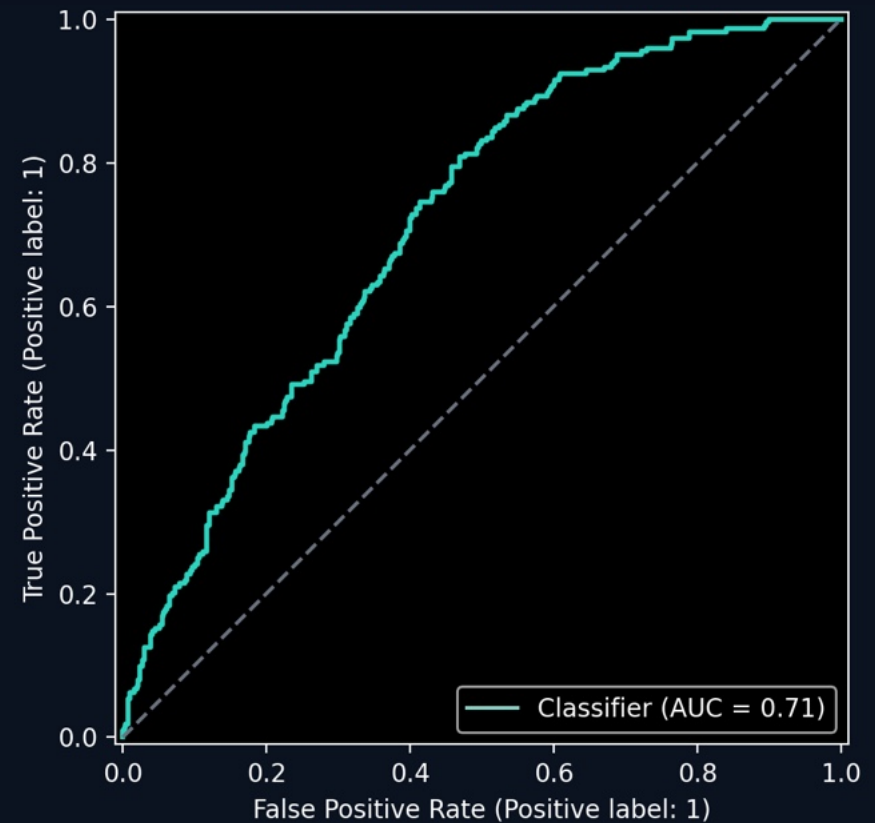


ROC Curve Analysis

There is a 71.4% probability of assigning a higher score to a thermal day versus a non-thermal one.

AUC=0.714 means that the model has real discriminatory capacity.

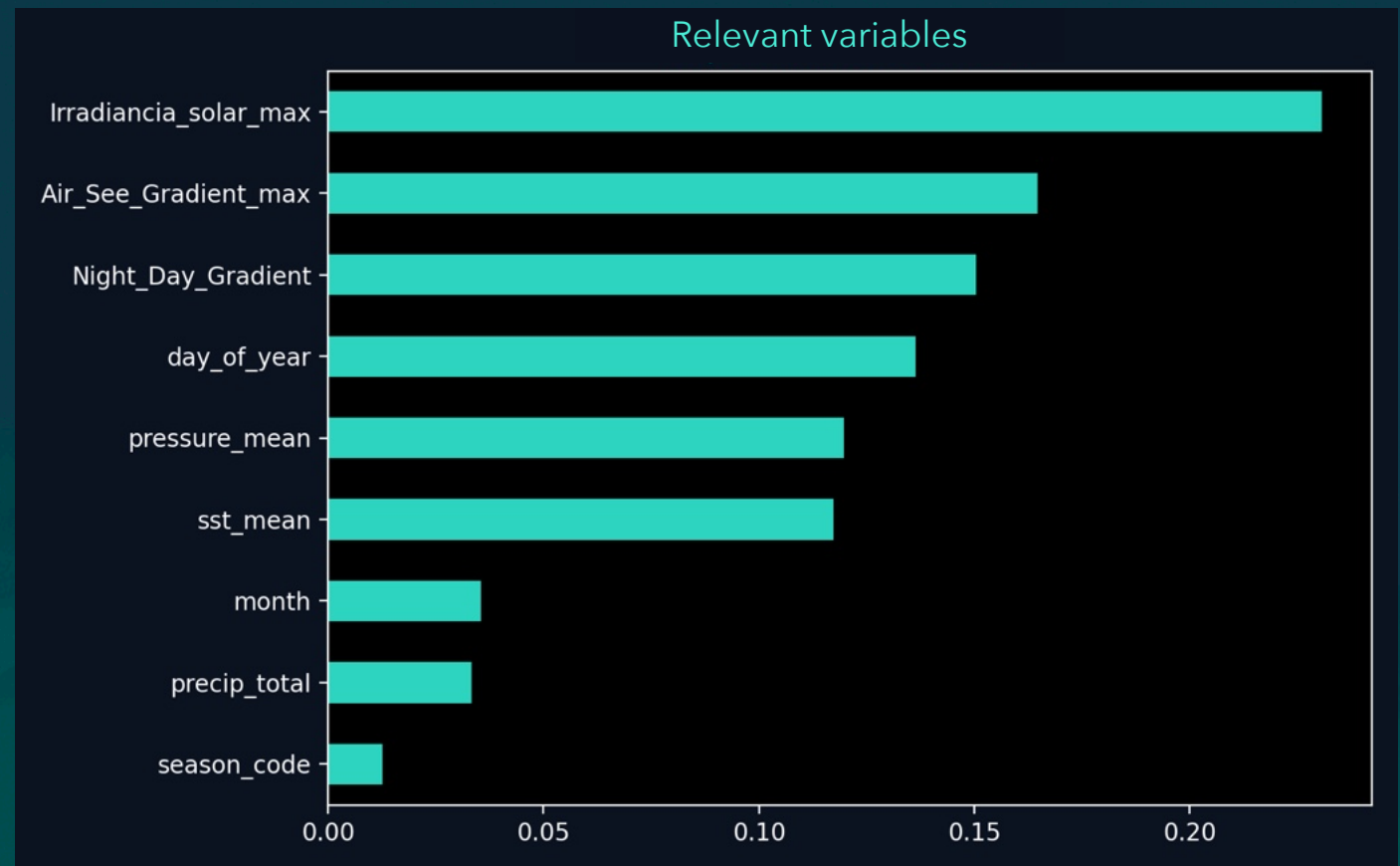
ROC Curve - Random Forest Model for Thermal Wind Prediction



Feature Importance

The model learns coherent physical mechanisms.

1. Irradiancia_solar_max (23.1%)
2. Air_See_Gradient_max (16.5%)
3. Night_Day_Gradient (15.0%)
4. day_of_year (13.6%)
5. pressure_mean (12.0%)
6. sst_mean (11.7%)



Validation Analysis

Proximity of results demonstrates temporary stability and absence of severe overadjustment.

Validation:

- F1 = 0.470

- AUC = 0.701

Test:

- F1 = 0.518

- AUC = 0.714

≡ RANDOM FOREST – VALIDATION ≡					
		precision	recall	f1-score	support
	0	0.822	0.621	0.708	536
	1	0.375	0.629	0.470	194
	accuracy			0.623	730
	macro avg	0.599	0.625	0.589	730
	weighted avg	0.703	0.623	0.645	730
	F1:	0.470			
	AUC:	0.701			

Limitations

- Variables are calculated using same-day information.
- The current model must be considered, mainly a historical classifier.
- The following evolution must incorporate variables available before 07:00 and external weather models.

Roadmap

- Current phase: Historical analysis and scientific validation.
- Future phase: GO/NO GO operating system
- Based on Open-Meteo, GFS and early daily prediction.
- Real application of short-term predictions in service and App. Buscaviento

Final Conclusions

The project demonstrates that Sea Breeze occurrence in Badalona is not random.

Machine Learning successfully captures meaningful atmospheric patterns and provides actionable decision support for water sports.

Test F1: 0.52
AUC test: 0.71
Accuracy (GO): 0.45
Recall (GO): 0.62
Historical positive days: 29.4%

- ✓ Real data · 16 years old · Badalona
- ✓ Confirmed physical hypothesis
- ✓ ML overcomes chance (AUC 0.71)
- Next step: operational forecast at 07:00

THANK YOU

“The best session is not the one that happens.
It is the one you know is going to happen.”

